

Tackling Group Theory computationally with GAP

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Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

Testing Conjectures

Finding Trends

Advanced Topics

Table of Contents

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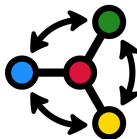
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Finding Trends

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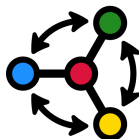
History

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- ▶ GAP was started at Lehrstuhl D für Mathematik, RWTH Aachen in 1986.



RWTHAACHEN
UNIVERSITY

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- ▶ It has a yearly conference (This year in Portugal)
- ▶ It also has an yearly excellence awards!

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- ▶ Provide a rich library of functions for algebraic computations.
- ▶ Support for programming and scripting to automate tasks.
- ▶ Extensive documentation (gap-system.org/doc) and a supportive community.

Why GAP & computation?

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- ▶ Number of groups can get BIG (non-isomorphic groups of order ≤ 2000 (423 million groups!)).
- ▶ Can and has helped uncover patterns not easily seen at first.

What we're going to do today?

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1. Given a group, explore what we can about it.
2. Prove/Disprove a claim about automorphism groups.
3. Conjecture trends about the commutativity ratio.

Online setup

We will be using mybinder to run GAP in an online Jupyter notebook. For a local setup, please refer to the GAP installation page, but we warn this might take time.

mybinder

Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

Testing Conjectures

Finding Trends

Advanced Topics

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- ▶ This makes it ideal for interactive exploration and rapid prototyping
- ▶ You can type commands directly and see results immediately

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- ▶ Comments start with #
- ▶ Assignment uses := (not =)
- ▶ Comparison uses = for equality
- ▶ Comparison uses <> for inequality

```
# This is a comment
```

```
x := 5;          # Assignment
```

```
y := x + 3;;    # No output
```

```
x = 5;          # Returns true
```

```
x <> 5;         # Returns false
```

Lists and Indexing

Lists are fundamental in GAP:

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- ▶ Indexing starts at **1** (not 0!)
- ▶ Access elements with `list[i]`
- ▶ List ranges: `[1..10]` creates `[1, 2, ..., 10]`

```
mylist := [10, 20, 30];  
mylist[1];          # Returns 10  
mylist[2];          # Returns 20  
primes := [2, 3, 5, 7, 11];
```

Control Flow Keywords

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- ▶ `if ... then ... elif ... else ... fi;`
- ▶ `for ... in ... do ... od;`
- ▶ `while ... do ... od;`
- ▶ `function (...) ... end;`

```
for i in [1..5] do
  if i mod 2 = 0 then
    Print(i, " is even\n");
  fi;
od;
```

Functions

Defining functions in GAP:

```
square := function(x)
  return x * x;
end;
```

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square(5);  # Returns 25
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Anonymous functions with `->`:

```
double := x -> 2 * x;
double(7);  # Returns 14
```

Your turn

Write a function that computes the n -th Fibonacci number.

Recall: The Fibonacci sequence is defined as:

- ▶ $F(0) = 0$
- ▶ $F(1) = 1$
- ▶ $F(n) = F(n-1) + F(n-2)$ for $n \geq 2$

Test your function with: `Fib(10)`

Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

Testing Conjectures

Finding Trends

Advanced Topics

Presentation of group

GAP provides built-in functions for a lot of known group types.
These include:

- ▶ $\mathbb{Z}_n$ as `CyclicGroup(n)`
- ▶ S_n as `SymmetricGroup(n)`
- ▶ A_n as `AlternatingGroup(n)`
- ▶ D_n as `DihedralGroup(2n)`
- ▶ Matrix Groups
- ▶ Quaternion Groups

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Furthermore, any arbitrary group's elements will be represented by permutations (why?).

Analyzing a group

GAP can answer problems about the group and its elements. It can:

- ▶ List all the elements - `Elements(G)`
- ▶ List all the subgroups - `AllSubgroups(G)`
- ▶ List all the normal subgroups - `NormalSubgroups(G)`
- ▶ Get the order of an element - `Order(g)`

Analyzing a group

GAP can also check if a certain property holds for a group by prepending the property with the `Is` noun:

- ▶ `IsAbelian`
- ▶ `IsCyclic`
- ▶ `IsNormal`
- ▶ `IsIsomorphic`

Your turn

Choose some group and find the following properties:

- ▶ Order
- ▶ Number of Conjugacy classes
- ▶ Order of the Largest Proper Normal Subgroup
- ▶ Center
- ▶ Exponent (google what that is)
- ▶ Minimal Generating set

Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

Testing Conjectures

Finding Trends

Advanced Topics

Testing Conjectures

GAP provides an easy setting to check statements and conjectures. In this section, we want to check whether the following is true or false.

Is the automorphism group of a non-abelian group non-abelian?

Morphisms

In GAP, a morphism is a function between two objects of the same type (category).

$$\text{Set} \rightarrow \text{Set}$$
$$\text{Grp} \rightarrow \text{Grp}$$
$$\text{Ring} \rightarrow \text{Ring}$$

Group Homomorphisms

For groups, we can define a homomorphism in two ways:

1. Defining the image of the generators.
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1. Defining the image of the generators.
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We also have `IsHomomorphism` to check if a function actually can be a homomorphism.

Group Automorphism

Automorphisms for groups follow the same rules as homomorphisms.

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We also have `AutomorphismGroup(G)` to get the automorphism group.

Searching All Groups

GAP maintains a library of all groups up to certain orders. You can search through them using:

- ▶ `AllSmallGroups( $n$ )` - Returns all groups of order n
- ▶ `NumberSmallGroups( $n$ )` - Returns the number of groups of order n
- ▶ `SmallGroup( $n$ ,  $i$ )` - Returns the i -th group of order n

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You can also filter by properties:

- ▶ `AllSmallGroups(Size, n, IsAbelian, true)`
- ▶ `AllSmallGroups(Size, n, IsCyclic, false)`

Your turn

Check our claim again.

Is the automorphism group of a non-abelian group non-abelian?

Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

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Finding Trends

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Finding Trends

GAP can also be great at analyzing trends and ideas.

Commutativity Ratio

For a finite group G the probability that two group elements chosen at random with replacement commute, which we denote by $\text{Pr}(G)$, is given by

$$\text{Pr}(G) = \frac{|\{(a, b) \in G \times G \mid ab = ba\}|}{|G|^2}$$

We call $\text{Pr}(G)$ the *commutativity ratio* of G .

Your turn

- ▶ Compute $\text{Pr}(G)$ for dihedral groups of order 6, 8, and 12
- ▶ Look for patterns in the results
- ▶ Can you conjecture a formula for $\text{Pr}(G)$ in terms of the number of conjugacy classes?

Your turn

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Hint: The number of conjugacy classes can be found using `NrConjugacyClasses(G)`. Try to use `SmallGroups` to help you.

Table of Contents

Introduction

GAP as a Programming Language

Looking at Groups in GAP

Testing Conjectures

Finding Trends

Advanced Topics

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GAP offers many advanced features:

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Free Presentations:

- ▶ Define groups by generators and relations using `FreeGroup`
- ▶ Example: $D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$

GAP in Research

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- ▶ Classification of groups of small order
- ▶ Character theory computations for sporadic simple groups
- ▶ Verification of algebraic structures in representation theory
- ▶ **Ore Conjecture (2010)**: Proved that every element of every finite non-abelian simple group is a commutator. GAP was used extensively for computational verification across thousands of groups.

Over 3000+ papers cite GAP in their research!

Further Exploration

Want to practice more with GAP?

- ▶ Joseph Gallian's *Contemporary Abstract Algebra* includes computer exercises designed for GAP

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- ▶ Your CURIOSITY!

Thank You!

Questions?

Resources: gap-system.org